

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		$2\text{Fe} + 3\text{Br}_2 \rightarrow 2\text{FeBr}_3$ award (1) for correct formulae of reactants award (1) for correct formula of product award (1) for correct balancing only if all formulae are correct		3		3		
		(ii)		add silver nitrate (solution) (1) iron(III) bromide gives a cream precipitate (1) iron(II) iodide gives a yellow precipitate (1) accept other sensible test e.g. add chlorine (water) (1) iron(III) bromide gives orange colour (1) iron(II) iodide gives brown/dark colour (1) add sodium hydroxide (solution) (1) iron(III) bromide gives a brown precipitate (1) iron(II) iodide gives a green precipitate (1)	2	1		3		3
	(b)			$\frac{19.4}{19} : \frac{16.3}{80}$ (1) $0.204 : 1.021$ (1) $1 : 5 \text{ ratio} \Rightarrow \text{BrF}_5$ (1) award (2) for correct answer with no working ecf possible award (2) for Br_5F (where A_r divided by mass with working shown) award (1) for Br_5F with no working		3		3	3	
				Question 11 total	2	7	0	9	3	3